SEAA 4333 REINFORCED CONCRETE DESIGN II · WEEK 1

Staircases I

Geometry, load on plan, and the stair as a beam

Workbook ch. 1 | ABCD card | Phone or laptop

Staircase Lab

Scan for this page’s Explorer tab. You will use it in four activities today.

HOW RCD2 RUNS

Four tools, one routine every week

01

Deck

Paces the session. No talk runs past ten minutes.

02

Workbook

Your module book. You write in it every class, in exam format.

03

Lab and app

Predict, then check. Feedback on every step of your working.

04

Studio

Fifteen minutes a week on your own project building.

Three habits

- Cards up when asked
- Phones for Lab windows only
- Hand calculation first, app second

THE SEMESTER · 12 OCT 2026 TO 24 JAN 2027

Five topics, fourteen Units

W1–2
Staircases
today

W3–6
Frame analysis
sub-frames, wind, sway

W7, 9–10
Columns
short, slender, biaxial

W11–13
Foundations
pads, straps, pile caps

W14–15
Retaining walls

Assessment	Weight	When
Test 1	15%	Week 9 (7–13 Dec), after the mid-semester break
Test 2	15%	Week 13 (4–10 Jan)
Project (group and individual)	15%	Submitted as the brief states; built in studio every week
Assignment	5%	As announced
Final exam	50%	31 Jan to 19 Feb 2027

TODAY · 180 MINUTES

How the next three hours run

0:00 **Welcome** How RCD2 runs	0:08 **Warm-up** RCD I in five votes	0:18 **Hook** Is this stair legal?	0:22 **Anatomy, rules** UBBL 106 and 108
0:33 **Stair Audit** Measure a real stair	0:53 **Stair to beam** Load on plan, span, M	1:08 **Faded example** In pairs	1:30 **Break** 10 minutes
1:40 **Checks, detailing** And the kink	1:53 **Spot the error** In teams	2:10 **Exam-size problem** Alone, 25 minutes	2:35 **Studio, exit** Your building's stairs

No talk longer than ten minutes. Rust times are your turn to work.

WARM-UP · RCD I RECALL · VOTE WITH YOUR CARD

Five questions on slab design, no notes

W1. A one-way slab has K = M/(bd²fck) = 0.090. Does it need compression steel?

A Yes, K > 0.05 · **B** No, K < Kbal = 0.167 · **C** Only if fyk = 500 · **D** Only if h > 200 mm

W2. Deflection of a simply supported one-way slab: the structural factor K in Table 7.4N is

A 1.0 · **B** 1.3 · **C** 1.5 · **D** 0.4

W3. Distribution (secondary) bars in a one-way slab must be at least

A 0.13% of bh · **B** 20% of the main steel · **C** 50% of the main steel · **D** the same as the main steel

WARM-UP · RCD I RECALL · VOTE WITH YOUR CARD

Two more

W4. The design action on a floor carrying g_k and q_k (EN 1990, Eq. 6.10) is

A $1.35g_k + 1.5q_k$ · **B** $1.4g_k + 1.6q_k$ · **C** $1.5g_k + 1.35q_k$ · **D** $g_k + q_k$

W5. In a slab no deeper than 200 mm, cracking is controlled by

A computing the crack width · **B** the bar spacing rules · **C** Table 7.2N for every bar · **D** nothing at all

Why this warm-up: a stair waist is a slab. Steps 7 to 11 of today's design are the slab design you did in RCD I.

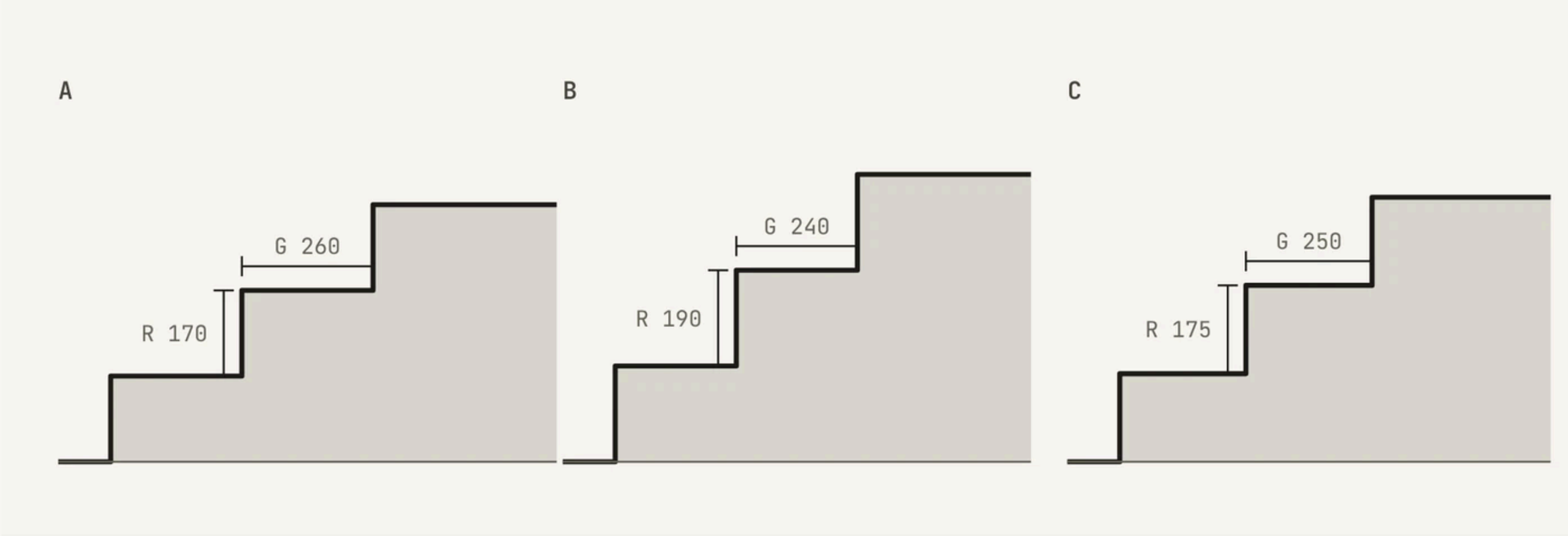
WARM-UP · ANSWERS

B, A, B, A, B, and what each means for a stair

Q	Answer	On a stair today
W1	B: K < Kbal = 0.167, singly reinforced	The waist uses the same K, z and As
W2	A: K = 1.0 when simply supported	Stairs use 1.0, 1.3 or 1.5 by their continuous ends
W3	B: 20% of the main steel	Same rule; we take 20% of the steel provided
W4	A: 1.35gk + 1.5qk	Same factors, with loads per m² on plan
W5	B: bar spacing rules	Same rules, tighter where the moment is largest

HOOK · PAIRS · 4 MINUTES

Which of these stairs is legal?



Decide with your neighbour: A, B, C, all of them, or none? Write it down. We check in ten minutes.

BLOCK A · THE METHOD

A stair is a slab that climbs

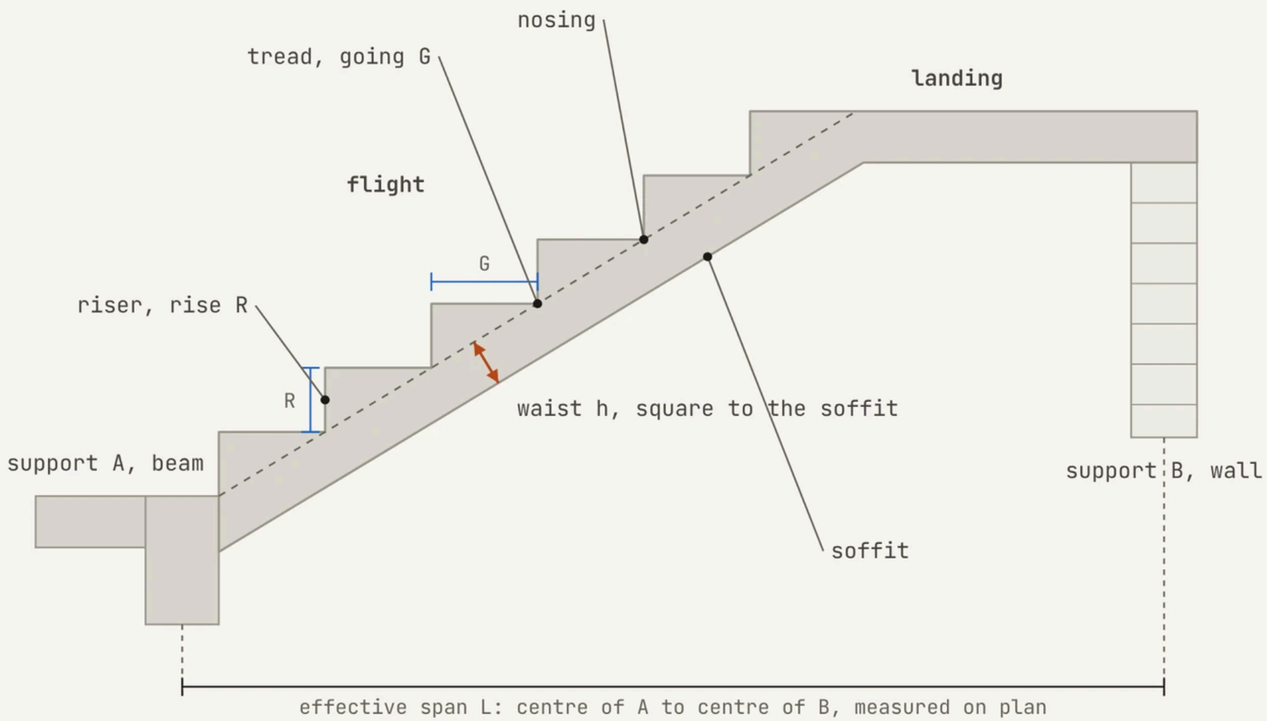


By 3:00 you can

- set R, G and the risers for a storey, and check them against UBBL 106 and 108
- turn a waist into a load per m^2 on plan
- model a flight, with or without its landing, as a beam on plan, and find F, M and the reactions
- design and detail the waist, and find the slips in someone else's calculation

BLOCK A · ANATOMY

The parts of a stair



Waist h is measured square to the soffit, not vertically.

The span is on plan, centre to centre of the supports.

This week: flights that span along their length, with or without landings.

BLOCK A · THE BY-LAWS

Four numbers fix the geometry

≤ 180

mm rise R, the same on every step

UBBL 106

≥ 255

mm tread G, the same on every step

UBBL 106

≤ 16

risers in one flight, then a landing

UBBL 108

≈ 600

mm for 2R + G, for comfort (550 to 700)

Course notes

Hook answer: only A is legal. C is comfortable (2R + G = 600) but its 250 mm tread is short of 255. B fails both the rise and the tread.

CONCEPT CHECK 1 · THINK 60 S, THEN VOTE

Our school has a 3.5 m storey. What is the least number of risers, and in how many flights?

A 19 risers, one flight

B 20 risers, two flights

C 20 risers, one flight

D 18 risers, two flights

CONCEPT CHECK 1 · ANSWER

B: 20 risers of 175 mm, in two flights of 10

Step 1

n ≥ 3500/180 = 19.4

so n = 20, rounded up

Step 2

R = 3500/20 = 175 mm

≤ 180, so legal

Step 3

20 > 16 per flight

two flights of 10, half landing at 1.75 m

Rounding down to 19 gives R = 184 mm: illegal. With G = 260, 2R + G = 610 mm. This is the stair Lab tab 1 opens with.

ACTIVITY 1A · TEAMS OF FOUR · 20 MINUTES

Stair Audit: is our stair legal?

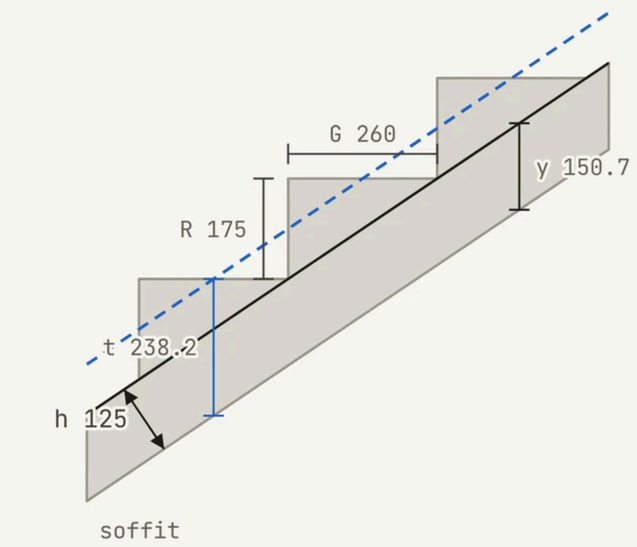
1. Walk to the stair your lecturer names. Measure five risers and five goings, and take the averages.
2. Count the risers in one flight. Measure the stair width and the landing depth.
3. Back in your seat: Lab tab 1, *Check a measured stair*. Record each verdict in Workbook 1.3.
4. The reporter adds your team to the class table and explains any fail.

Measurer | Recorder | Checker | Reporter

Back in your seats by 0:46 · class table at 0:50

BLOCK B · LOAD ON PLAN

From waist to load on plan



$$y = h\sqrt{(R^2 + G^2)}/G$$

$$t = y + R/2$$

$$gk = 25t + finishes$$

$$nd = 1.35gk + 1.5qk$$

Why on plan? The span is measured on plan, and people stand on the treads. WE 1.1: h 125, R 175, G 260 → y 150.7, t 238.2 mm, self-weight 5.95 kN/m².

ACTIVITY 1B · CONCEPT CHECK 2 · PREDICT, THEN REVEAL · PAIRS · 5 MIN

Same waist (150 mm) and going (260 mm). R rises from 150 to 180 mm. How much does the self-weight on plan rise?

A 0%: the waist did not change

B About 10%

C About 20%, like R

D About 40%

Vote in Workbook 1.4 · then reveal in Lab tab 2

ACTIVITY 1B · ANSWER

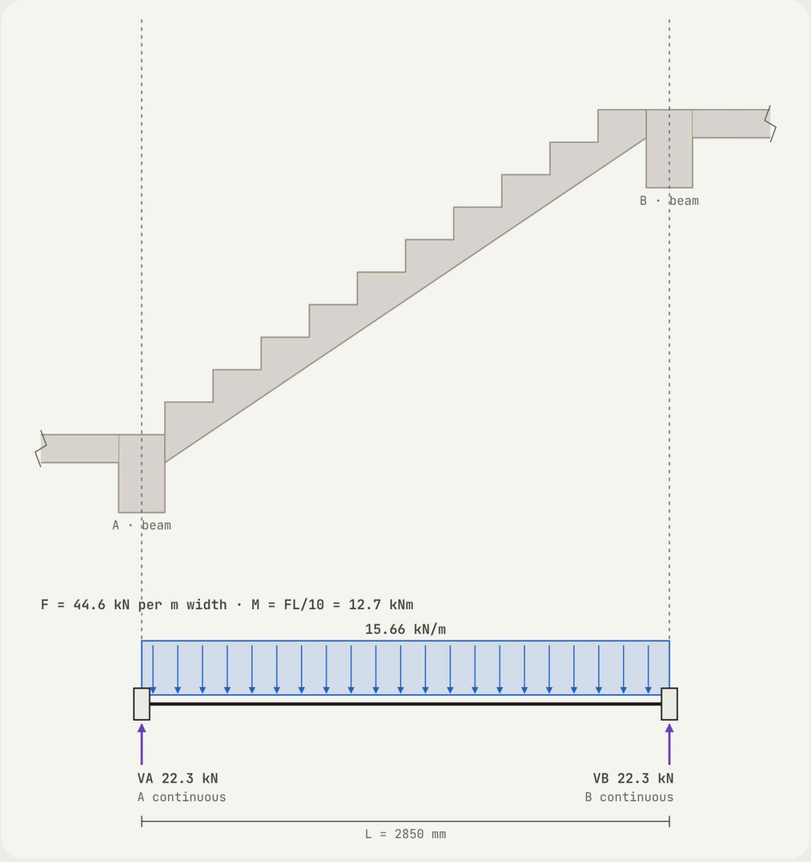
B: about 10%, from 6.20 to 6.81 kN/m²

R (mm)	y (mm)	R/2 (mm)	t (mm)	25t (kN/m²)
150	173.2	75	248.2	6.20
180	182.4	90	272.4	6.81
change	+5%	+20%	+10%	+10%

R rose 20%, but most of t is the waist y, and y grew only 5%. The steps add R/2.

BLOCK B · FROM STAIR TO BEAM

Model the stair as a beam on plan



Span L: centre to centre of the supports, on plan.

F = Σ nd × length, per metre width: the flight part and the landing part.

M = FL/10 with at least one continuous end; **FL/8** when both ends are simply supported.

Deflection K (Table 7.4N): 1.5 both ends continuous, 1.3 one end, 1.0 neither.

BLOCK B · WORKED EXAMPLE 1.1 · I DO

Flight 1 of our 3.5 m storey, between two beams

R 175 · **G** 260 mm

10 × 260 = 2600 mm between beam faces

Beams 250 mm wide, both continuous

Waist 125 mm; cover 25; H10

gk 1.2 finishes; **qk** 4.0 kN/m²

C25/30; fyk 500

Calculation	Output
y = 125√(175² + 260²)/260	150.7 mm
t = 150.7 + 175/2	238.2 mm
gk = 25 × 0.2382 + 1.2	7.15 kN/m²
nd = 1.35 × 7.15 + 1.5 × 4.0	15.66 kN/m²
L = 2600 + 250/2 + 250/2	2850 mm
F = 15.66 × 2.85	44.6 kN
M = FL/10 = 44.6 × 2.85/10	12.7 kNm
RCD I slab design, d = 95 mm	**A**s 325 → H10-225

BLOCK B · THE DESIGN PROCEDURE

Eleven steps; three are new

1 Design life, exposure, fire

2 Materials

3 Waist h, then average thickness t

4 Cover: durability, fire, bond

5 Actions per m^2 on plan

6 Span, F, M = FL/10, reactions

7 Flexure: K, z, As (RDF §6)

8 Shear (RDF §7)

9 Deflection, K by the ends (RDF §8)

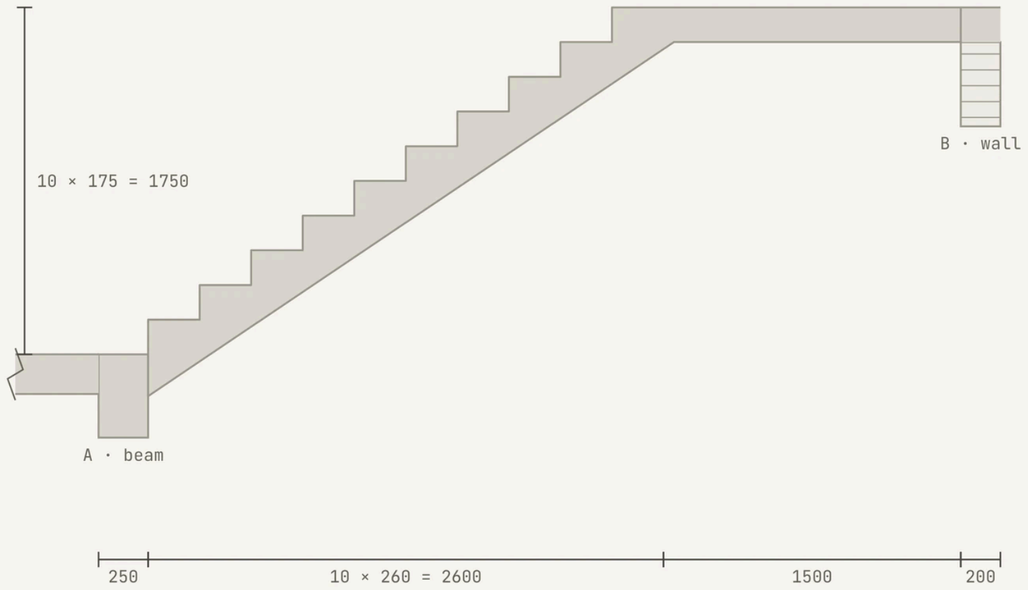
10 Cracking: bar spacing (RDF §9)

11 Detailing: top bars, the kink

Dark rows are new for stairs. Steps 7 to 11 are RCD I slab design, with the stair's own d, K and spacing limits.

ACTIVITY 1C · PAIRS · 22 MINUTES

Faded Example 1.2



R 175, G 260; waist and landing 175 mm

A: beam 250, continuous · B: wall 200, simple

gk 1.2 finishes, qk 4.0 kN/m²; C25/30; H10

Fill the blank lines in Workbook 1.6 by hand. Then pick **FE 1.2** in Lab tab 3 and check L, F, M and the reactions.

Check at 1:25 · answers at 1:27

ACTIVITY 1C · ANSWERS

Check: L = 4325 mm, M = 30.2 kNm

t 298.4 mm	nd flight 17.69 kN/m²	nd landing 13.53 kN/m²	L = 2725 + 1600 4325 mm
F 69.9 kN	M = FL/10 30.2 kNm	VA, continuous end 44.0 kN	VB, wall 25.8 kN

Main steel, d = 145 mm

As 506 → H10-150 (524)

Deflection, K = 1.3

35.4 allowed ≥ 29.8 actual

1:30

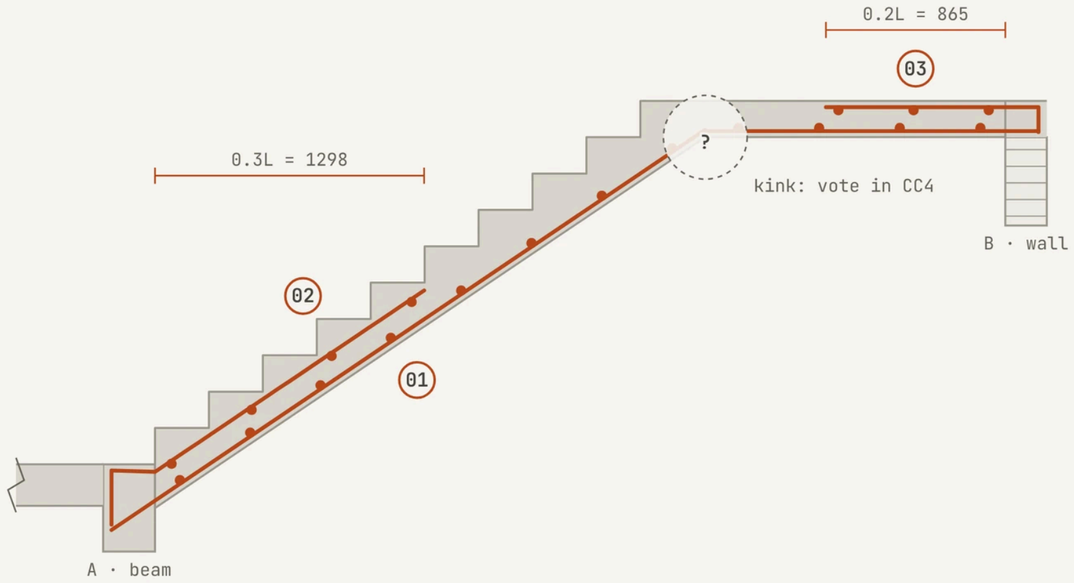
Break

10 minutes. Back at 1:40.

Next: the checks stairs get wrong, and a calculation with four slips in it.

BLOCK C · CHECKS AND DETAILING

What a stair adds to the slab checks



Shear: the continuous end carries more; take ρl from the top bars there.

Deflection: K = 1.5, 1.3 or 1.0 by the continuous ends.

Spacing where M peaks: main ≤ 2h and 250; distribution ≤ 3h and 400 mm.

Distribution steel: at least 20% of the main steel provided.

Top bars: 0.3L at a continuous end; at least 25% of the span steel over 0.2L at a simple end.

CONCEPT CHECK 3 · THINK 60 S, THEN VOTE

A flight with its landing spans from a monolithic beam (A) to a brick wall (B). Which pair is right?

A M = FL/8 and K = 1.0

B M = FL/10 and K = 1.3

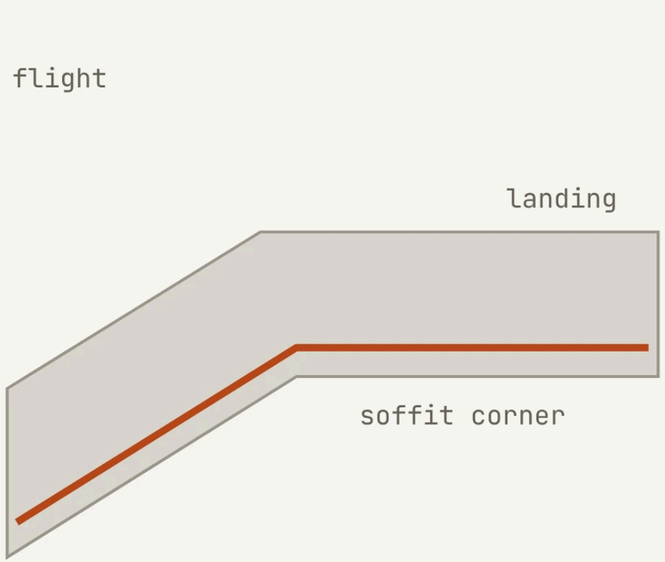
C M = FL/10 and K = 1.5

D M = FL/12 and K = 1.5

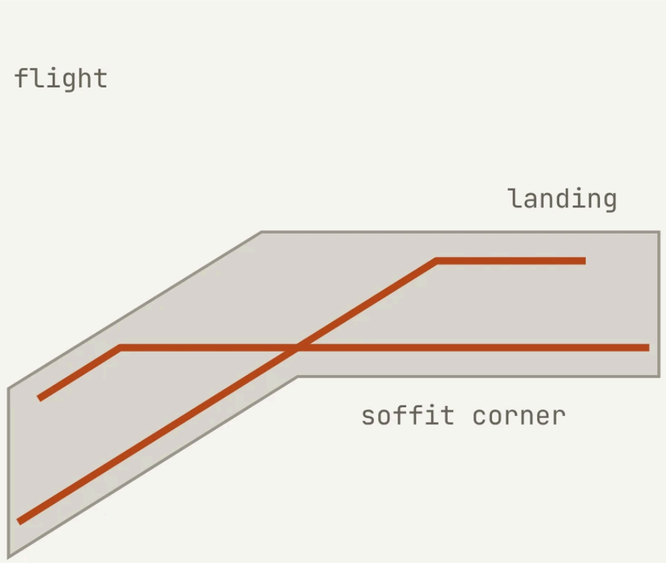
CONCEPT CHECK 4 · THINK 60 S, THEN VOTE

The flight meets its top landing. Which bottom-bar detail is right at the kink?

A



B



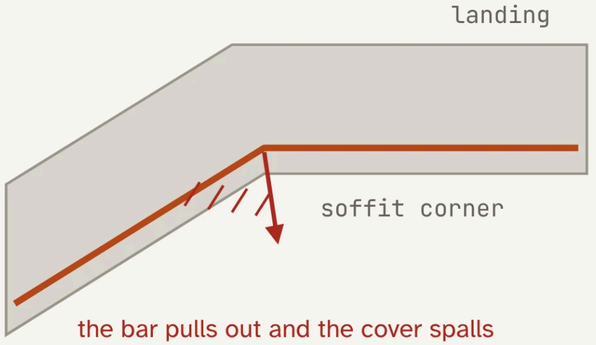
Vote · convince your neighbour · vote again

B and B

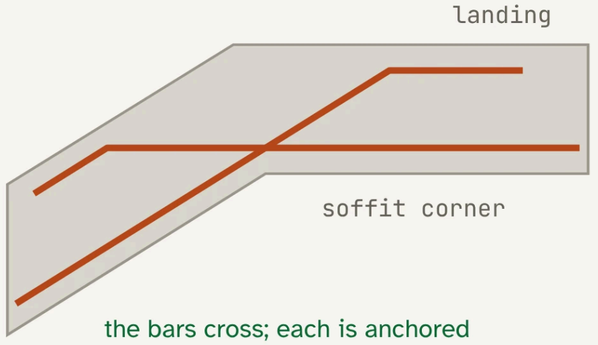
CC3 · B

One continuous end: M = FL/10, and the end-span factor K = 1.3.

Both ends continuous would give K = 1.5; neither, FL/8 and K = 1.0.



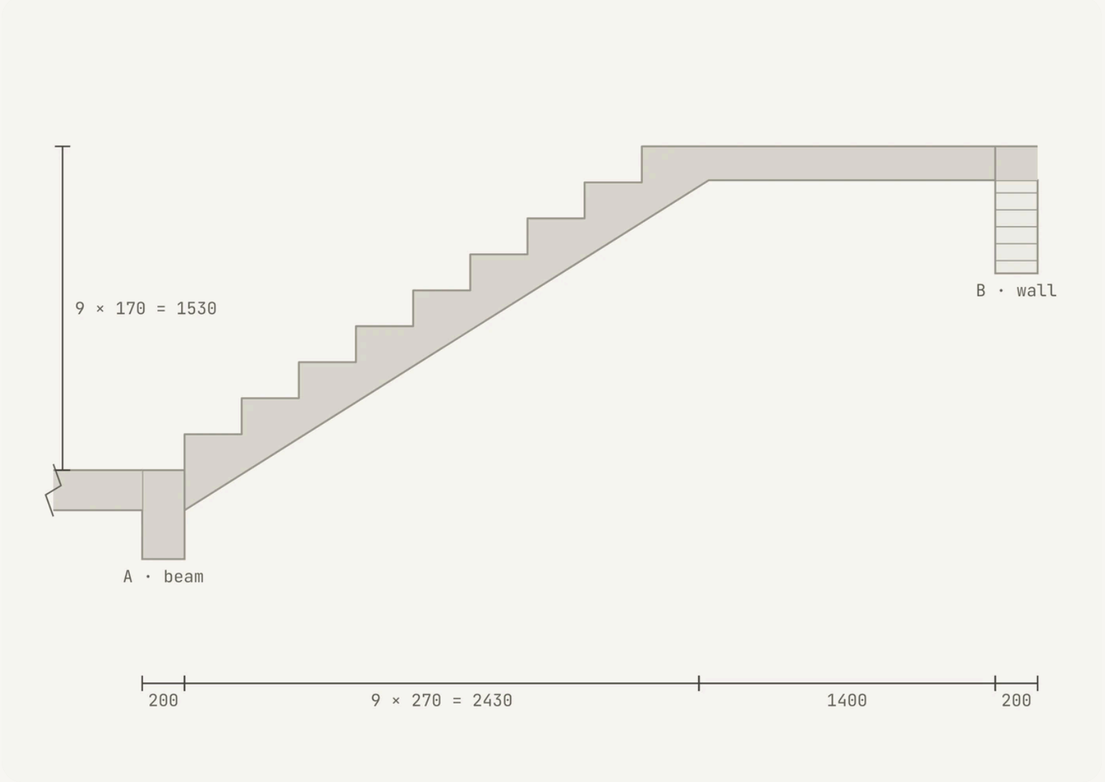
A, wrong: the bar straightens and spalls the corner.



B, right: the bars cross and anchor in the compression zone.

ACTIVITY 1D · TEAMS OF FOUR · 17 MINUTES

Stair S1 has four slips. Find them.



Workbook 1.8 holds the submitted calculation: fifteen lines in Ref / Calculations / Output.

1. Alone, 5 min: mark every line you doubt.
2. Team, 8 min: agree on four, with a one-line reason each.
3. Lab tab 5: tap your lines, press Check.

Roles: reader, calculator, checker, reporter

ACTIVITY 1D · ANSWERS

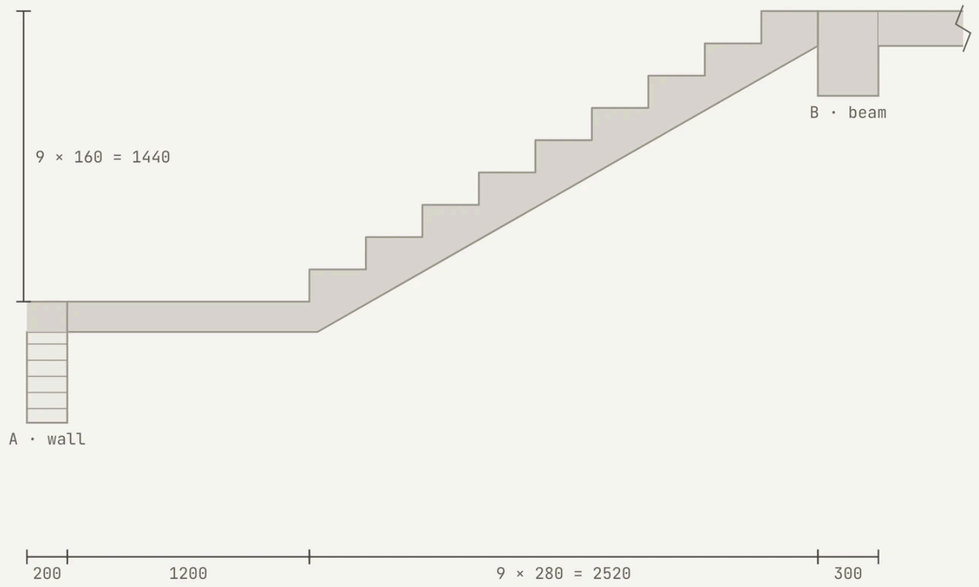
Four small slips, one unsafe stair

Slip	Line	What went wrong	Correct
A	4	Self-weight from the waist h	From t: 25 × 0.274 = 6.85 kN/m²
B	6, 7	1.35 on the imposed load	1.5: nd = 16.60 and 12.75 kN/m²
C	8	Clear span, 3830 mm	Centre to centre: 4030 mm
D	13	K = 1.5 with one continuous end	K = 1.3 (end span)

Result: M = 17.8 kNm instead of 24.6. H10-225 gives 349 mm²/m where 461 is needed: 24% short. Correct design: H10-150.

BLOCK D · EXAM-SIZE PROBLEM · ALONE · 25 MINUTES

D1: a flight with a bottom landing



A: 200 mm wall, simply supported. B: 300 mm beam, continuous. Waist and landing 150 mm. Finishes 1.5, imposed 4.0 kN/m². C30/37, fyk 500, cover 25 mm, H10.

Part	Marks
(a) Average thickness and design loads	6
(b) Span and design moment	4
(c) Main and distribution bars	8
(d) Deflection check, after class	7

Ref / Calculations / Output. No Lab, no app.

PROJECT STUDIO · PROJECT GROUPS · 17 MINUTES

Propose the stairs for your building

1. Take each storey height from your brief (in 2024/25: 3.5 m, then 3.0 m).
2. Choose n, R and G. Check UBBL 106 and 108 and 2R + G in Lab tab 1.
3. Choose the arrangement: flights between beams, or a flight with its landing.
4. Take a first waist of span/25 to span/30, using Lab tab 3 for the span.
5. Log it in Workbook 1.10, with a sketch of the plan.

Deliverable today

One line per storey: n × R, G, flights, waist, arrangement.

It starts your group's staircase design for the project.

CLOSE · 2:52 · PROFESSIONAL HABIT 1

Sanity-check the load before you design

Stair	nd flight	nd landing	Ratio
WE 1.1	15.66	11.84	1.32
FE 1.2	17.69	13.53	1.31
S1 as submitted	12.15	12.15	1.00

A flight carries more per m² on plan than its landing.

A ratio of 1.0 or less means a slip. In S1 it catches slip A; slip B lowers both loads alike, so check the partial factors too.

AFTER CLASS

Before Week 2

Finish

D1 part (d), then self-mark all of D1 with the answer scheme.

Explore

RC Design app, Staircase tab: Ex 8.1(a) and 8.1(b), step by step.

Read

The staircase chapter of Yassin and Abdullah (2022), Examples 8.1 to 8.3.

Bring

Workbook chapter 1 and your ABCD card.

Week 2: stairs supported by landings, two flights at right angles, and stairs with their sides built into walls.

EXIT TICKET · 5 MINUTES · WORKBOOK 1.11

Before you leave

1. A stair has R = 175 mm and G = 250 mm. Is it legal under By-law 106? Why?

2. Waist h = 150 mm, R = 175 mm, G = 260 mm. Find the average thickness t.

3. What is still unclear for you today?

Tear off the slip and hand it in at the door. Next week: stairs on landings.